

virtio-blk scalability improvements

[KVM on IBM Z and LinuxONE: virtio-blk scalability improvements](#)

qemu支持virtio-blk的多队列读写。

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QEMU 9.0 and libvirt 10.0 introduced an important improvement for disk (virtio-blk) scalability. Until now, a virtio-blk device could use one **iothread** in the host (or the QEMU main thread).

For a while it was already possible to specify multiple virtio queues for a single virtio-block device, and the Linux guest driver was able to exploit these concurrently:

```
<driver name='qemu' queues='3'>
```

All queues have been handled by one **iothread**. This introduced a limit for the maximum amount of IO requests per second per disk no matter how many queues have been defined.

Now it is possible to assign a queue to a given host **iothread** allowing for higher throughput per device:

```
<driver name='qemu' queues='3'>
  <iothreads>
    <iothread id='2'>
      <queue id='1' />
    </iothread>
    <iothread id='3'>
```

[Improve virtio-blk device performance using iothread-vq-mapping](#)

 [Scaling virtio-blk disk I/O with IOThread Virtqueue Mapping | Red Hat Developer](#)

This article covers the IOThread Virtqueue Mapping feature for Kernel-based virtual machine (KVM) guests that was introduced in [Red Hat Enterprise Linux \(RHEL\) 9.4](#).

The problem

Modern storage evolved to keep pace with growing numbers of CPUs by providing multiple queues through which I/O requests can be submitted. This allows CPUs to submit I/O requests and handle completion interrupts locally. The result is good performance and scalability on machines with many CPUs.

Although virtio-blk devices in KVM guests have multiple queues by default, they do not take advantage of multi-queue on the host. I/O requests from all queues are processed in a single thread on the host for guests with the `<driver io=native ...>` libvirt domain XML setting. This single thread can become a bottleneck for I/O bound workloads.

KVM guests can now benefit from multiple host threads for a single device through the new IOThread Virtqueue Mapping feature. This improves I/O performance for workloads where the single thread is a bottleneck. Guests with



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